GROUP ACTIVITY / LIVE KPI DASHBOARD

JPMorgan Chase & Co.

KPI Rationale and Strategic Interpretation

Thesis: JPMorgan is converting a record technology budget into digital customer scale and operating leverage, so that growth no longer requires proportional growth in branches, headcount, or cost.

Member	Role	Contribution statement
Shahzaib Ali	Member A Metric Selector	Selected JPMorgan Chase and defined the seven-metric set, locking metric names across all three deliverables. Researched the public disclosures behind each metric, then wrote the KPI Rationale and the Strategic Interpretation.
Nazifa Maliha	Member B Data Analyst	Sourced, audited, and reconciled historical data for all seven locked metrics across 2017–2025 using JPMorgan Chase SEC Form 10-K filings, CCB earnings supplements, Investor Day presentations, and J.D. Power national studies. Documented four primary inflection points with operational causes and audited the Strategic Interpretation against the disclosures to guarantee 100% internal data consistency.
Roopa Shankar Rao	Member C Dashboard Builder	Built the live HTML and Chart.js dashboard from the verified data package, including the KPI cards, charts and source table. Highlighted the 2022–2023 inflection points in the charts so the visuals match the written interpretation.

1. KPI Rationale

Seven locked metrics: 4 leading, 3 lagging. Word count: 419.

Selection logic. JPMorgan Chase's thesis is that sustained technology investment converts into digital customer scale and operating leverage, so growth no longer requires proportional growth in cost or headcount. We therefore chose seven metrics that follow the causal chain from inputs, to customer behavior, to financial outcomes, rather than seven independent measures. Each answers a question the others cannot. Four are leading and three are lagging.

Leading indicators. *Technology Expense as % of Net Revenue* measures commitment. It shows whether the firm is investing more intensively or simply growing with revenue, and it moves years before efficiency does. We use the firm's own "total technology expense" series, because earlier "tech budget" figures use a narrower definition. *Active Digital Customers* (users logged in within 90 days) tests whether investment builds adoption; the definition is stable across the whole window. *Mobile Share of Digital Customers* (mobile actives divided by digital actives) separates depth from breadth: a rising share shows migration to the primary, lowest-cost and most data-rich channel. *Chase Branch Count* is deliberately a test of the thesis, not a confirmation of it. If digital substitutes for physical distribution, branches should fall; a rise signals a hybrid model. It is leading because branch openings are capacity investments that precede customer and deposit growth by years.

Lagging indicators. *Overhead Ratio* (noninterest expense divided by net revenue, reported basis) is the headline test of operating leverage. *Net Revenue per Employee* adds what the ratio cannot: productivity that is independent of wage inflation, showing whether output scales faster than headcount. *J.D. Power Satisfaction Gap vs. Study Average* checks that efficiency is not bought at the customer's expense. We plot Chase's gap to the average rather than its raw score, so industry-wide movements are not mistaken for firm-specific performance.

Why the set is diagnostic. Overhead ratio alone cannot distinguish genuine efficiency from a revenue windfall; pairing it with tech intensity and revenue per employee can. Digital customers alone cannot show whether adoption pays; pairing them with branch count and satisfaction can. We omitted the platform dimension because JPMorgan's thesis is not a marketplace thesis, and we excluded stock price (market expectations, not progress) and AI use-case counts (too few comparable periods).

Known confounders. Net interest income makes revenue rate-sensitive: net revenue rose 23% in 2023, so the overhead ratio and revenue per employee partly reflect the rate cycle. The May 2023 First Republic acquisition inflates 2023 headcount and branches. A $7.9B Visa share gain lifted 2024 revenue. The dashboard footnotes each where it occurs.

2. Strategic Interpretation

Written from the dashboard's data and annotated inflection points. Word count: 324.

> **JPMorgan's dashboard supports a narrower claim than its transformation thesis: technology investment has sustained the firm's efficiency gains, but the metrics do not show it creating them.**

Start with what did not change. Active Digital Customers grew roughly 5% to 7% every year from 2017 to 2025, reaching 74.6 million, with no acceleration after technology expense rose from $11.7B in 2019 to about $18B in 2025. Mobile Share of Digital Customers climbed from 64% to 83%, but annual gains shrank from over three points to one, a sign of saturation. Technology Expense as % of Net Revenue stayed between 9.6% and 11.0% in every disclosed year, so spending tracked the business rather than moving ahead of it.

Now what did change. The Overhead Ratio sat at 59% in 2021 and 2022, then fell to 52% by 2024. Net Revenue per Employee dipped to $438K in 2022 as headcount grew 6% and then 8%, before rising to $573K by 2025. Both inflections coincide with a 23% jump in 2023 net revenue, reflecting higher interest rates and the First Republic acquisition, and not with any inflection in digital adoption. From 2022 to 2025 revenue rose about 42% against expense growth of about 26%; that gap, not technology intensity, accounts for the seven-point improvement.

Chase Branch Count is the most revealing signal. It fell from 5,130 in 2017 to a trough of 4,787 in 2022, then rose to 5,083 while digital customers grew 60%. Digital and physical channels behave as complements. Meanwhile Chase's J.D. Power gap over the study average narrowed from 30 points in 2022 to 11 in 2025, so digital scale has not produced a growing satisfaction premium.

The strategic question is whether the 2026 technology budget of $19.8B and its AI programs can bend Technology Expense as % of Net Revenue and Net Revenue per Employee once rates normalize. Until those lines move independently of revenue, the transformation has proven resilience, not yet leverage.

Appendix | Supporting Material

Not part of the word counts. Gives Members B and C the locked metric contract, the working dataset, and the proposed inflection points.

A. Locked metric contract (names must match in the dashboard, Rationale, Interpretation and sources)

#	Metric name	Type	Definition	Chart
1	**Technology Expense as % of Net Revenue**	Leading	Firmwide total technology expense divided by reported total net revenue	Line
2	**Active Digital Customers**	Leading	CCB users of web and/or mobile who logged in within 90 days (millions)	Line
3	**Mobile Share of Digital Customers**	Leading	Active mobile customers divided by active digital customers (%)	Line / bar
4	**Chase Branch Count**	Leading	Year-end CCB branches	Bar
5	**Overhead Ratio**	Lagging	Noninterest expense divided by net revenue, reported basis (%)	Line
6	**Net Revenue per Employee**	Lagging	Reported net revenue divided by year-end headcount ($ thousands)	Bar
7	**J.D. Power Satisfaction Gap vs. Study Average**	Lagging	Chase score minus national-bank study average, 1,000-pt scale (points)	Bar

The donut or stacked-bar requirement can be met by showing the Mobile Share of Digital Customers metric as a 2025 mobile versus non-mobile donut beside the line.

B. Working dataset (calendar year-end; verify every figure against the cited filing before use)

Metric	2017	2018	2019	2020	2021	2022	2023	2024	2025
Tech expense ($B)	n/d	n/d	11.7	n/d	12.5	14.1	15.3	17.0	~18.0 est.
Net revenue, reported ($B)	100.7	109.0	115.6	120.0	121.6	128.7	158.1	177.6	182.4
1 Tech expense % of revenue	n/d	n/d	10.1	n/d	10.3	11.0	9.7	9.6	~9.9 est.
2 Active digital (M)	46.7	49.3	52.4	55.3	58.9	63.1	67.0	70.8	74.6
3 Mobile share of digital (%)	64.4	67.5	71.2	74.0	77.2	78.7	80.4	81.7	82.7
4 Branch count	5,130	5,036	4,976	4,908	4,790	4,787	4,897	4,966	5,083
5 Overhead ratio (%)	59	58	57	56	59	59	55	52	52
Headcount (000s)	252.5	256.1	257.0	255.4	271.0	293.7	309.9	317.2	318.5
6 Net revenue / employee ($K)	399	426	450	470	449	438	510	560	573
Chase J.D. Power score	n/a	n/a	n/a	n/a	n/a	678	674	677	677
Study average	n/a	n/a	n/a	n/a	n/a	648	653	658*	666
7 Satisfaction gap (pts)	n/a	n/a	n/a	n/a	n/a	+30	+21	+19	+11

n/d = not disclosed in the consistent series; n/a = study not collected on a comparable basis. Revenue per employee = reported net revenue / year-end headcount (calculated). Mobile share and J.D. Power gap are calculated. *2024 study average derived from the 2025 release (666, described as 8 points above 2024). 2025 technology expense is management's May 2025 outlook and must be labeled "est." on the dashboard. 2024 revenue includes a $7.9B Visa gain. Headcount for 2017 to 2019 comes from a secondary compilation; confirm against the 10-K.

C. Proposed inflection points for the dashboard annotations (Member B to confirm)

Chart	Period	What changed	Likely cause (one sentence)
Overhead Ratio	2022 to 2023	Fell from 59% (2022) to 55% (2023) and 52% (2024).	Net revenue jumped 23% in 2023 while noninterest expense grew about 14%, on higher interest rates and the First Republic acquisition.
Chase Branch Count	2022 to 2023	Decline from 5,130 (2017) reversed at a 4,787 trough, then +110, +69 and +117 in 2023 to 2025.	The 400-new-branch commitment was completed in 2022, followed by First Republic and continued expansion (more than 860 new branches since 2019 per the 2024 shareholder letter).
Tech Expense % of Revenue	2022 to 2023	Peaked at 11.0% in 2022 and fell to 9.7% in 2023.	Revenue grew 23% in 2023 while technology expense grew about 9%, so intensity fell without any cut in spending.
Net Revenue per Employee	2020 to 2022	Declined from $470K to $438K, then rose to $510K in 2023.	Headcount rose 6% in 2021 and 8% in 2022 ahead of revenue, then revenue outran hiring in 2023.

D. Sources (all figures traceable to these)

- JPMorgan Chase Form 10-K, FY2019 (selected metrics 2017 to 2019; five-year financial summary), FY2020, FY2022 and FY2024 (filed Feb 14, 2025).
- JPMorgan Chase Earnings Release Financial Supplements: 4Q20, 1Q22, 4Q25 and 1Q26 (branches, active digital and mobile customers).
- JPMorgan Chase Annual Reports 2019, 2022 and 2025, Financial Highlights (net revenue, noninterest expense, employees); 2022 and 2024 line-of-business shareholder letters.
- JPMorgan Chase Investor Day presentations: 2022 and 2023 (Global Technology), May 20, 2024 (transcript and deck), and 2025 (total technology expense series and outlook).
- J.D. Power U.S. National Banking Satisfaction Study press releases: 2022, Dec 14, 2023, Dec 12, 2024 and Dec 11, 2025.
- jpmorgan.com/technology (2026 technology investment of $19.8B).

E. Open items before submission

- Find technology expense for 2020 (and 2017 to 2018 if disclosed on the same basis); the series currently has six periods, enough for the four-period rule.
- The 2022 deck shows 2022 technology expense of $14.1B while the 2023 deck charts $14.3B, and the 2024 deck implies 2023 of about $15.5B versus $15.3B. Pick one basis and footnote it.
- Replace the 2025 estimate with the actual figure once confirmed in the 2025 annual report.
- Confirm the 2023 revenue-driver wording (rates and First Republic) against the 2023 Form 10-K MD&A before the Interpretation is final.
- J.D. Power redefined "national bank" over time (deposit threshold of $200B in 2022, $250B in 2025); footnote this on the satisfaction chart.